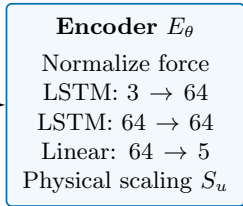


Used during allocation

Auxiliary training branch

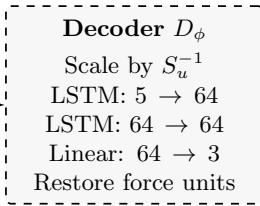
Desired force
 $\tau_k^d \in \mathbb{R}^3$



**Physical
commands**

$$\hat{\mathbf{u}}_k = \begin{bmatrix} \hat{F}_1 \\ \hat{F}_2 \\ \hat{\alpha}_2 \\ \hat{F}_3 \\ \hat{\alpha}_3 \end{bmatrix}$$

$\mathbb{R}^5$: N and rad



Learned force
 $\hat{\tau}_k \in \mathbb{R}^3$

$$\downarrow$$
$$\mathcal{L}_1(\tau^d, \hat{\tau})$$

Allocation model

$$g(\hat{\mathbf{u}}_k) = B(\hat{\alpha}_k) \hat{F}_k$$

Physical force
 $\tau_k^{\text{cmd}} \in \mathbb{R}^3$

$$\downarrow$$
$$\mathcal{L}_0(\tau^d, \tau^{\text{cmd}})$$

$\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5$
Magnitude, rate, power, sectors